

Sustech 2025南科大入学考试物理样卷-更新

NO. 1

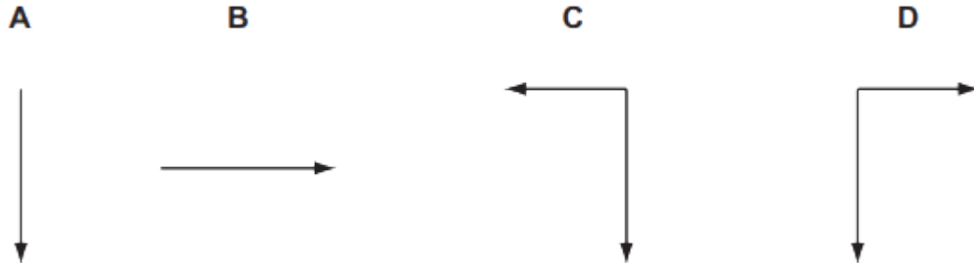
Which estimate is realistic?

- A. The kinetic energy of a bus travelling on an expressway is 30000J.
- B. The power of a domestic light is 300W.
- C. The temperature of a hot oven is 300K.
- D. The volume of air in a car tyre is 0.03m^3 .

NO. 2

An object in air is thrown upwards and towards the left.

Which diagram shows the force(s) acting on the body when it is at its highest point?



NO. 3

When a constant braking force is applied to a vehicle moving at speed v , the distance d moved by the vehicle in coming to rest is given by the expression

$$d = kv^2$$

where k is a constant.

When d is measured in metres and v is measured in metres per second, the constant has a value of k_1 .

What is the value of the constant when the distance is measured in metres, and the speed is measured in kilometres per hour?

- A. $0.0772 k_1$
- B. $0.278 k_1$
- C. $3.60 k_1$
- D. $13.0 k_1$

NO. 4

Which statement is correct?

- A. A ball lands on the ground and bounces. The kinetic energy changes sign, because the ball changes direction.
- B. A car drives up a slope at a steady speed. The power generated by the engine equals the potential energy gained per unit time.
- C. An electric heater can be 100% efficient.
- D. It is impossible for momentum to be conserved in a collision.

NO. 5

Two spheres approach each other along the same straight line. Their speeds are u_1 and u_2 before they collide. After the collision, the spheres separate with speeds v_1 and v_2 in the directions shown below.



The collision is perfectly elastic. Which equation must be correct?

- A. $u_1 - u_2 = v_2 + v_1$
- B. $u_1 - u_2 = v_2 - v_1$
- C. $u_1 + u_2 = v_2 + v_1$
- D. $u_1 + u_2 = v_2 - v_1$

NO. 6

The density of air on the Earth decreases almost linearly with a height from 1.22kgm^{-3} at sea level to 0.74 kg m^{-3} at an altitude of 5000m. Atmospheric pressure at the Earth's surface on a particular day is 100000Pa. The value of g between the Earth's surface and an altitude of 5000 m can be considered to have a constant value of 9.7ms^{-2} .

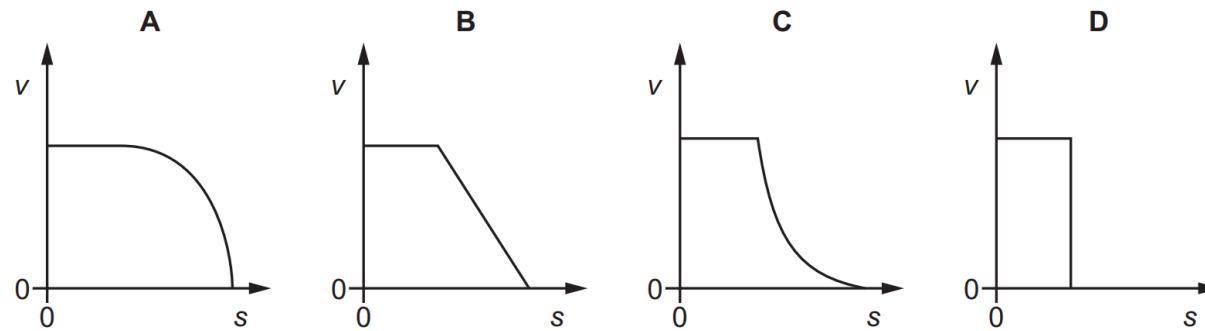
What will be the atmospheric pressure at an altitude of 5000m?

- A. 36000Pa
- B. 48000Pa
- C. 52000Pa
- D. 59000Pa

NO. 7

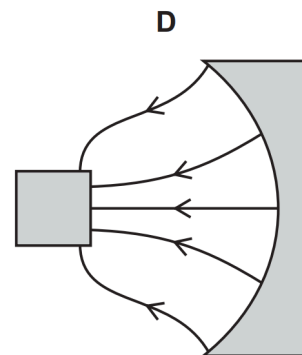
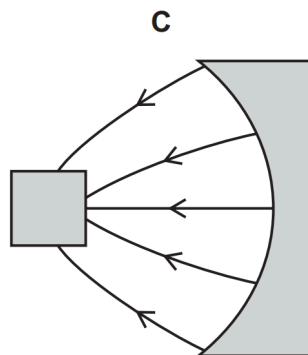
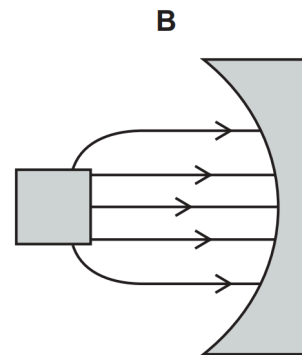
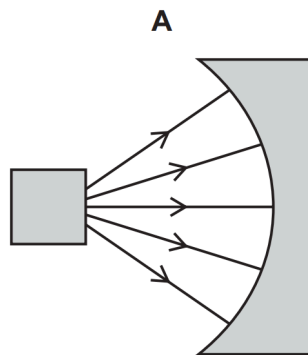
A car is travelling at constant velocity. Its brakes are then applied, causing uniform deceleration.

Which graph shows the variation with distance s of the velocity v of the car?



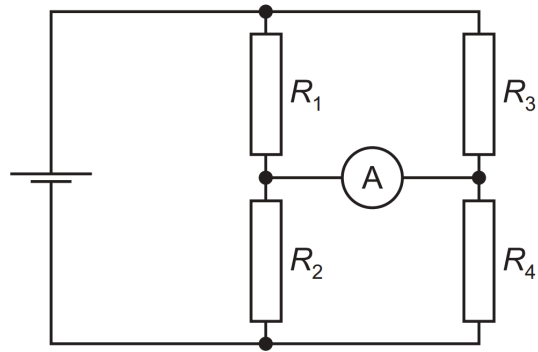
NO. 8

Which diagram could represent the electric field lines between two oppositely charged conducting surfaces?



NO. 9

In the circuit shown, the reading on the ammeter is zero.



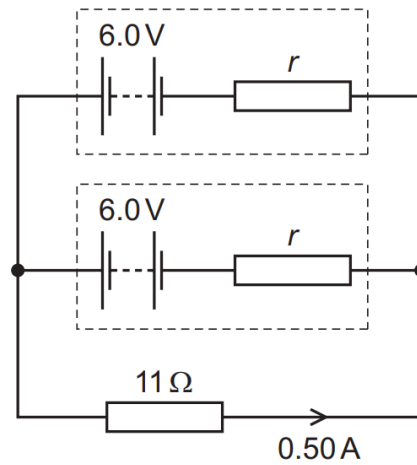
The four resistors have different resistances R_1 , R_2 , R_3 and R_4 .

Which equation is correct?

- A. $R_1 - R_3 = R_2 - R_4$
- B. $R_1 \times R_3 = R_2 \times R_4$
- C. $R_1 - R_4 = R_2 - R_3$
- D. $R_1 \times R_4 = R_2 \times R_3$

NO. 10

Two identical batteries each have e.m.f. 6.0V and internal resistance r . The batteries are connected to an external resistor of resistance 11Ω , as shown.



The current in the external resistor is 0.50A .

What is the internal resistance r of each battery?

- A. $1.0\ \Omega$
- B. $2.0\ \Omega$
- C. $4.0\ \Omega$
- D. $6.5\ \Omega$

NO. 11

Coherent light passes through a double slit, producing bright and dark fringes on a screen placed parallel to the plane of the double slit. The intensity of the light from each of the slits is initially the same.

The intensity of the light passing through one of the slits in the double slit is now increased. The frequency of the light remains constant.

What is the effect on the appearance of the fringes on the screen?

	separation of fringes	the maximum intensity of dark fringes
A	decreases	no change
B	increases	greater
C	no change	greater
D	no change	no change

NO. 12

Before a thunderstorm, the hairs on your head sometimes stand on end.

A hair with mass 0.50mg and charge 1.0pC is supported by a force due to an electric field. Ignore any forces other than the weight of the hair and the electric force.

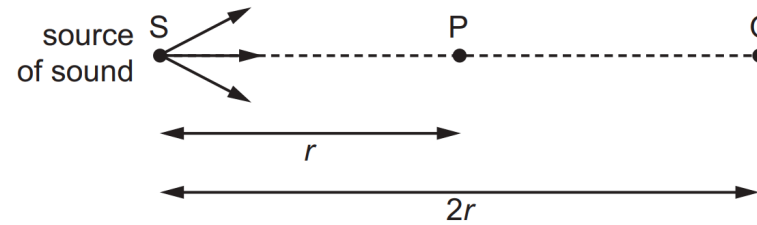
What is the electric field strength?

- A. $4.9 \times 10^3 \text{ NC}^{-1}$
- B. $4.9 \times 10^5 \text{ NC}^{-1}$
- C. $4.9 \times 10^6 \text{ NC}^{-1}$
- D. $4.9 \times 10^9 \text{ NC}^{-1}$

NO. 13

The intensity I of sound is inversely proportional to the square of the distance x from the source of the sound. This can be represented as

$$I \propto \frac{1}{x^2}.$$



Air molecules at point P, a distance r from the source S, oscillate with amplitude $8.0\mu\text{m}$.

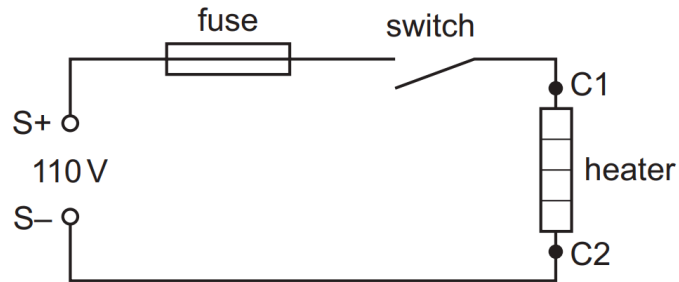
Point Q is situated a distance $2r$ from S.

What is the amplitude of oscillation of air molecules at Q?

- A. $1.4\mu\text{m}$
- B. $2.0\mu\text{m}$
- C. $2.8\mu\text{m}$
- D. $4.0\mu\text{m}$

NO. 14

A 110V supply of negligible internal resistance is connected to a heater through a fuse and a switch.



Terminals S+ and S– are the positive and negative terminals of the supply. Points C1 and C2 at either side of the heater are accessible for fault-finding.

A voltmeter is connected between S– and C1.

With the circuit working correctly, the voltmeter reading is noted with the switch closed.

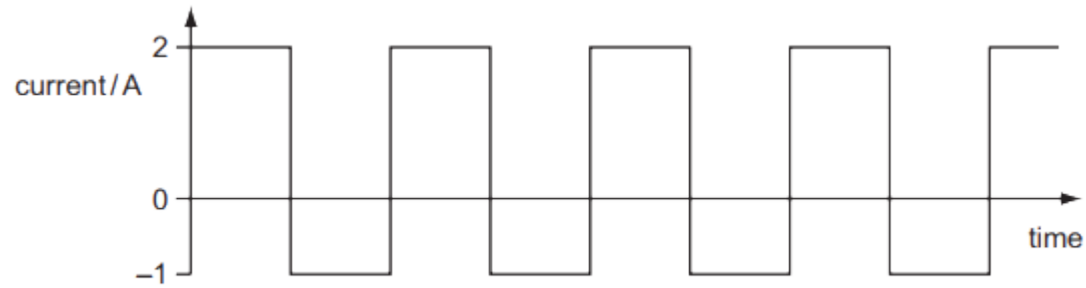
A fault occurs and the voltmeter is again connected between S– and C1 with the switch closed.

Which fault would result in the same two voltmeter readings?

- A a break in the wire of the heater
- B a broken switch that cannot close correctly
- C a melted fuse
- D a short circuit in the heater

NO. 15

A 100Ω resistor conducts a current with changing direction and magnitude, as shown.



What is the mean power dissipated in the resistor?

- A. 100 W
- B. 150 W
- C. 250 W
- D. 400 W

NO. 16

- a. A progressive longitudinal wave travels through a medium from left to right. Fig. 4.1 shows the positions of some of the particles of the medium at time t_0 and a graph showing the particle displacements at the same time t_0 .

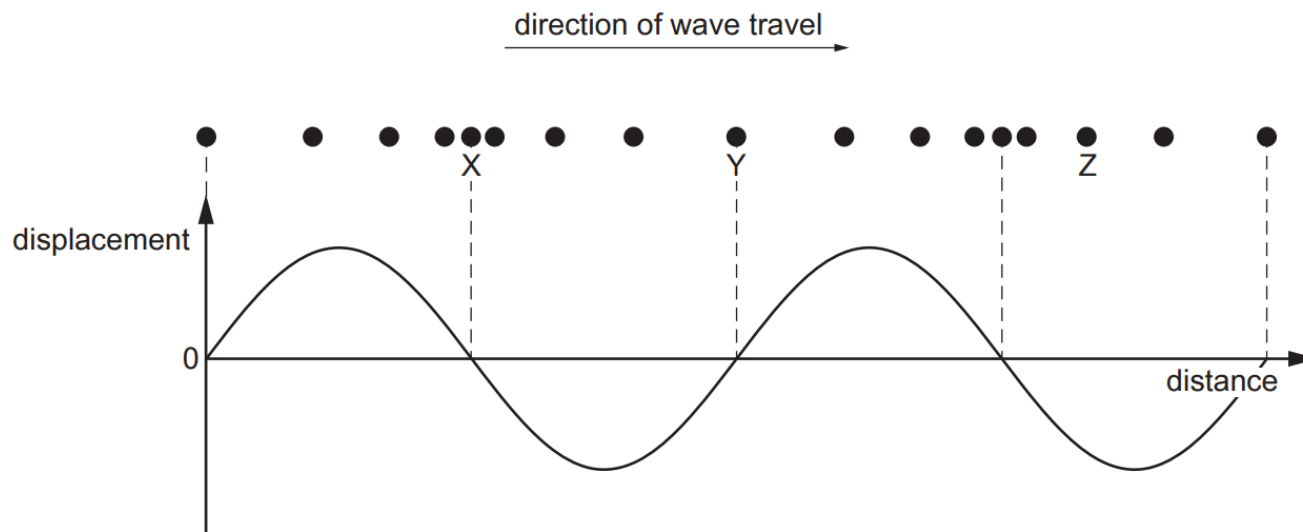


Fig. 4.1

Particle displacements to the right of their equilibrium positions are shown as positive on the graph and particle displacements to the left are shown as negative on the graph.

The period of the wave is T .

- On Fig. 4.1, draw circles around two particles which are exactly one wavelength apart. [1]
 - On Fig. 4.1, sketch a line on the graph to represent the displacements of the particles for the longitudinal wave at time $t_0 + \frac{T}{4}$. [3]
 - State the direction of motion of particle Z at time $t_0 + \frac{T}{4}$.
..... [1]
- b. The frequency of the wave in (a) is 16kHz. The distance between particles X and Y is 0.19m. Calculate the speed of the wave as it travels through the medium.

speed = ms^{-1} [3]

- c. A longitudinal sound wave is travelling through a solid. The initial intensity of the wave is I_0 . The frequency of the wave remains constant and the amplitude falls to half of its original value. Determine, in terms of I_0 , the final intensity of the wave.

intensity = I_0 [2]

- d. The sound wave in (c) now meets another sound wave travelling in the opposite direction.
i. State a condition necessary for these two waves to form a stationary wave.

..... [1]

- ii. State **two** ways in which a stationary wave differs from a progressive wave.

1

.....

2

.....

[2]

[Total: 13]

NO. 17

a. With reference to the photoelectric effect, state what is meant by the threshold *frequency*.

.....[2]

b. Electromagnetic radiation of wavelength λ is incident on a metal surface. Electrons of maximum kinetic energy E_{MAX} are emitted.

i. On Fig. 11.1, sketch the variation with $1/\lambda$ of E_{MAX} .

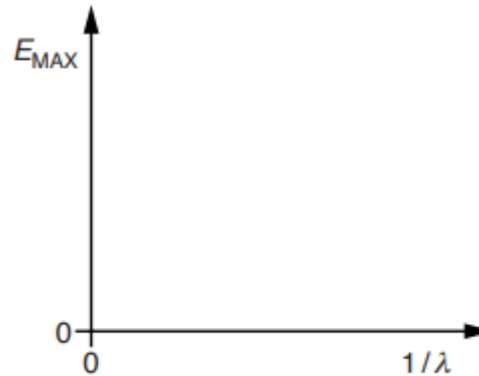


Fig. 11.1

[2]

ii. State an equation relating the gradient of the graph drawn on Fig. 11.1 to the Planck constant h .

Explain any symbols you use.

.....[1]

iii. Explain why, for any particular wavelength of electromagnetic radiation, most of the electrons are emitted with kinetic energies less than the maximum value E_{MAX} .

.....[2]

iv. Light of a particular wavelength is incident on a metal surface and gives rise to a photoelectric current.

The wavelength is reduced. The intensity of the light is kept constant.

State and explain the effect, if any, on the photoelectric current.

.....[3]

[Total: 10]

NO. 18

Two solid metal spheres A and B, each of radius 1.5cm, are situated in a vacuum. Their centres are separated by a distance of 20.0cm, as shown in Fig. 6.1.

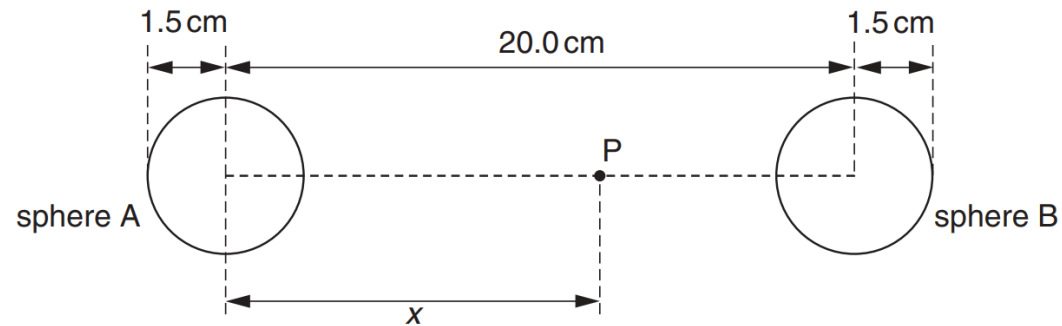


Fig. 6.1 (not to scale)

Both spheres are positively charged.

Point P lies on the line joining the centres of the two spheres, at a distance x from the centre of sphere A.

The variation with distance x of the electric field strength E at point P is shown in Fig. 6.2.

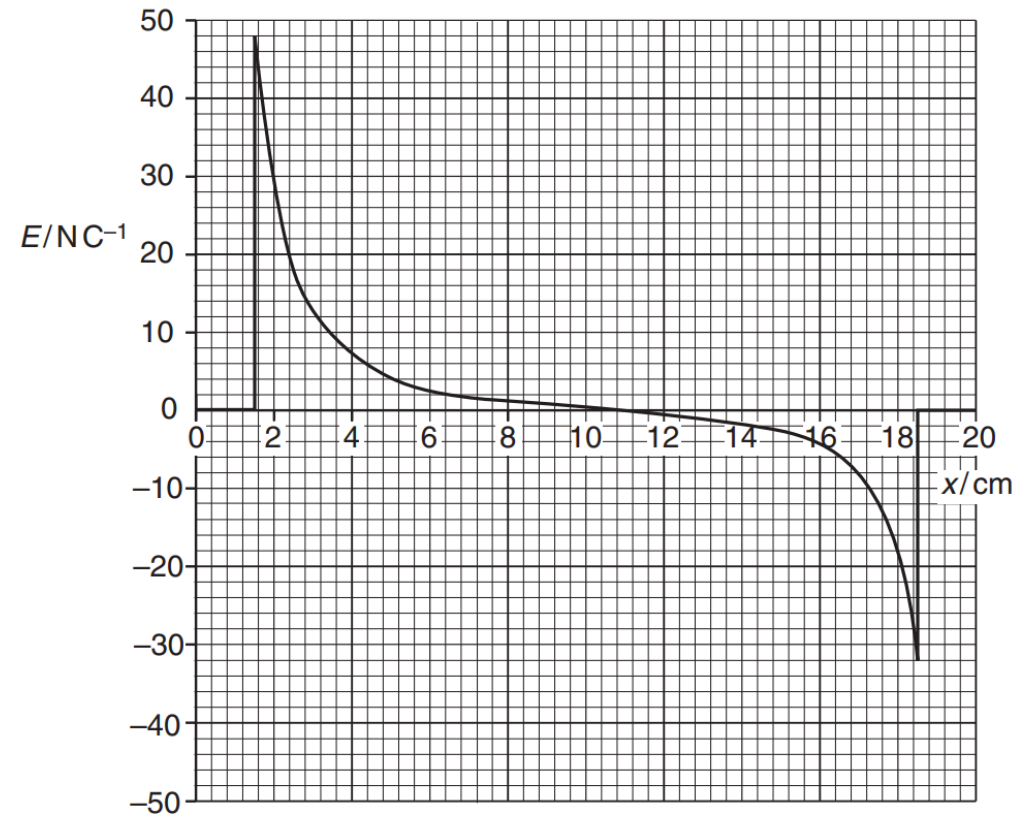


Fig. 6.2

a. Use Fig. 6.2 to determine the ratio

$$\frac{\text{magnitude of charge on sphere A}}{\text{magnitude of charge on sphere B}}.$$

Explain your working.

ratio =[3]

b. The variation with distance x of the electric potential V at point P is shown in Fig. 6.3.

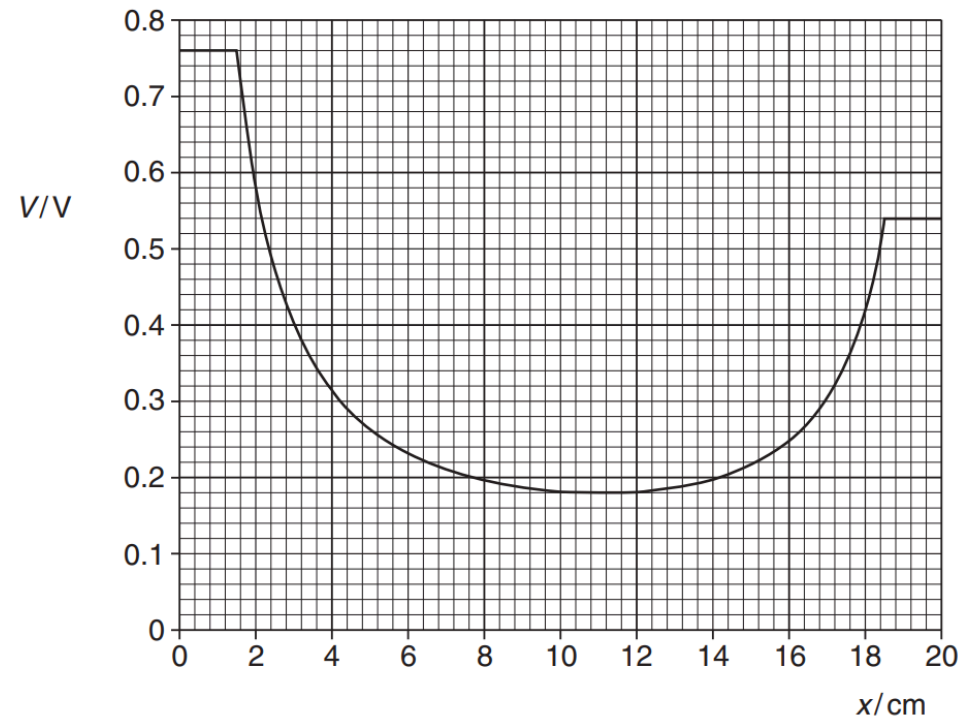


Fig. 6.3

An α -particle is initially at rest on the surface of sphere A.
The α -particle moves along the line joining the centres of the two spheres.

Determine, for the α -particle as it moves between the two spheres,
i. its maximum speed,

maximum speed = ms^{-1} [3]

ii. its speed on reaching the surface of sphere B.

speed = ms^{-1} [2]

[Total: 8]

NO. 19

A mass hangs vertically from a fixed point by means of a spring, as shown in Fig. 4.1.

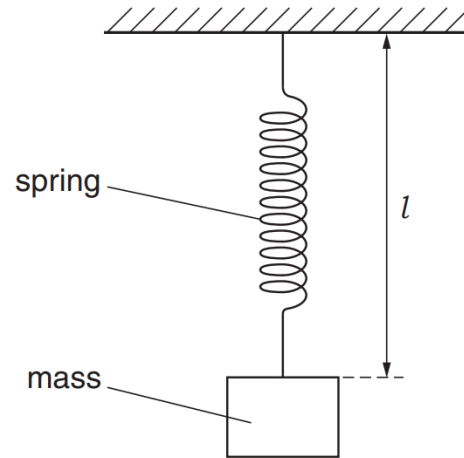


Fig. 4.1

The mass is displaced vertically and then released. The subsequent oscillations of the mass are simple harmonic. The variation with time t of the length l of the spring is shown in Fig. 4.2.

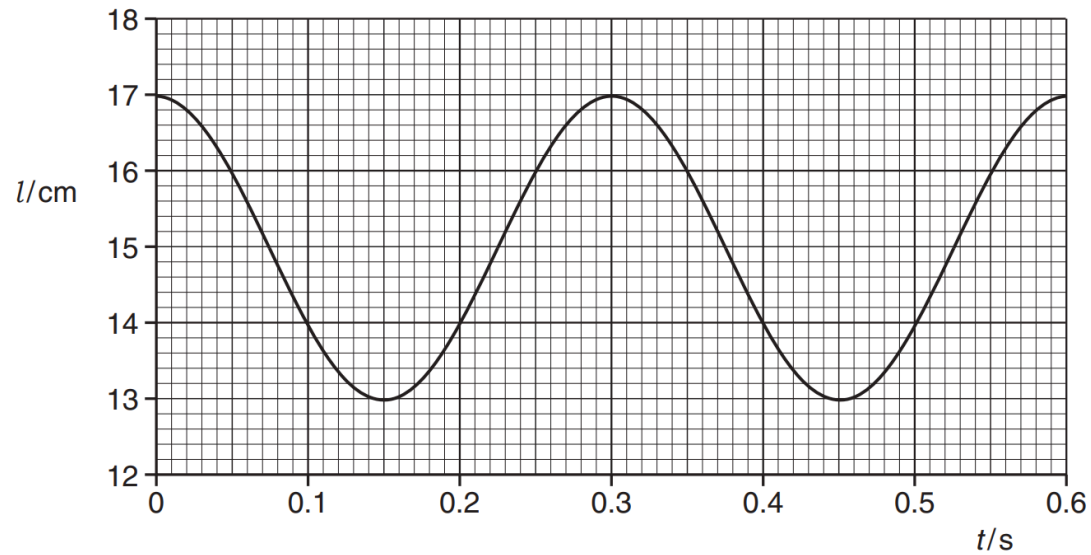


Fig. 4.2

a. Use Fig. 4.2 to

i. state two values of t at which the mass is moving downwards with maximum speed,

$t = \dots\dots\dots$ s and $t = \dots\dots\dots$ s [1]

ii. determine, for these oscillations, the angular frequency ω ,

$\omega = \dots\dots\dots$ rads^{-1} [2]

iii. show that the maximum speed of the mass is 0.42 ms^{-1} . [2]

- b. Use data from Fig. 4.2 and **(a)(iii)** to sketch, on the axes of Fig. 4.3, the variation with displacement x from the equilibrium position of the velocity v of the mass.

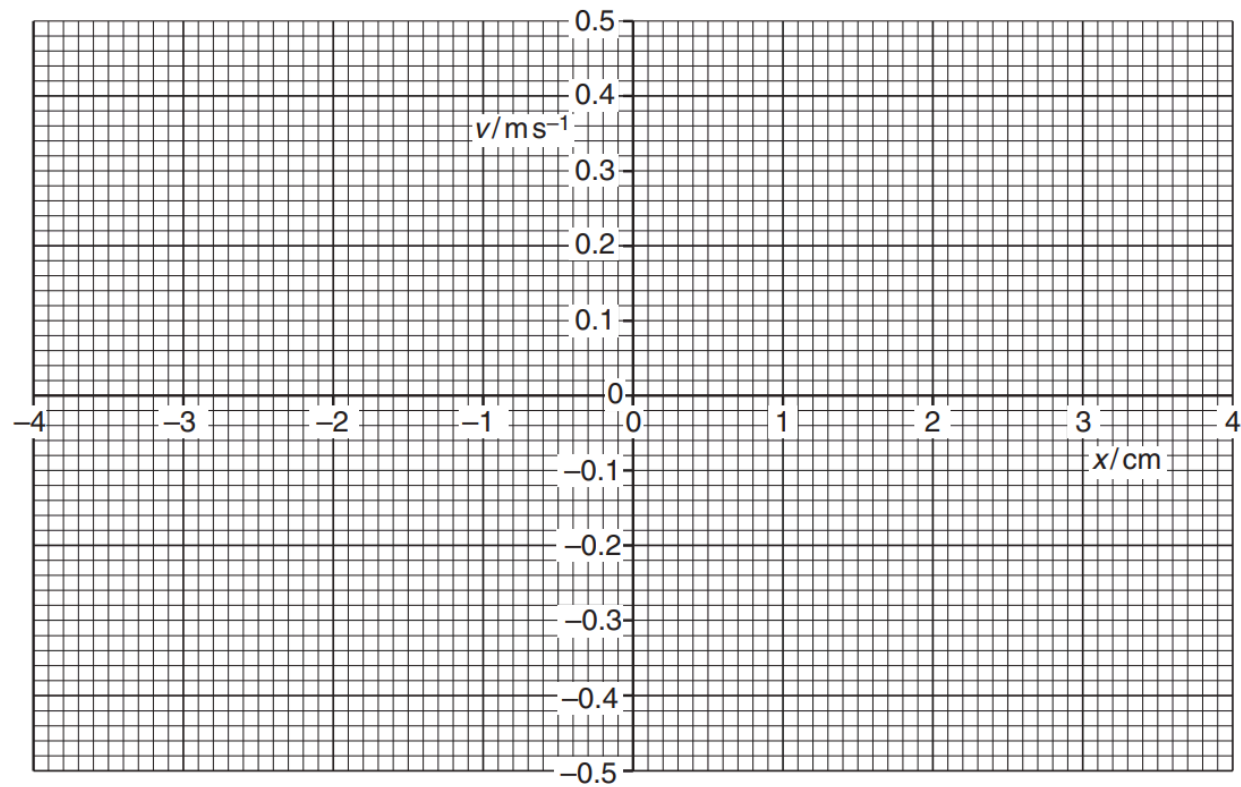


Fig. 4.3

[3]

[Total: 8]

